Session 20 Quiz - AI Bias & Fairness

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:57539]
AI Bias occurs when an AI system works better for some groups than others because it learned from ___ or incomplete data.

A. perfect
B. unfair
C. random
D. encrypted

Correct Answer: B. unfair

Explanation: AI systems reflect the patterns found in their training material. If that material is missing certain groups or contains historical prejudices, the AI learns those biases. Unfair data leads to unfair AI.

Q2 [Type: MCQ | Difficulty: Easy-Medium | QID:57540]
When an AI's training data does not include a diverse enough range of people, it is known as ___ bias.

A. Algorithm
B. Speed
C. Data
D. Software

Correct Answer: C. Data

Explanation: Data bias happens specifically because the information fed into the pattern finder is missing examples of certain groups. Algorithm bias comes from the math, and speed or software are not bias types.

Q3 [Type: MCQ | Difficulty: Easy | QID:57541]
The computer science rule '___ In, Garbage Out' means that if you train AI on biased data, it will produce biased results.

A. Truth
B. Code
C. Money
D. Garbage

Correct Answer: D. Garbage

Explanation: This phrase highlights that the quality of the output is entirely dependent on the quality of the input. Bad input (garbage) leads to bad output regardless of how good the algorithm is.

Q4 [Type: MCQ | Difficulty: Easy-Medium | QID:57543]
If a hiring AI was trained on past data where mostly men were hired, it might incorrectly learn that '___ = good hires.'

A. Computers
B. Men
C. Robots
D. Students

Correct Answer: B. Men

Explanation: The AI finds patterns in the history it is given. If men were the primary group hired in the past, it assumes that being male is a requirement for success, even if that is false.

Q5 [Type: MCQ | Difficulty: Easy-Medium | QID:57544]
Building ___ teams helps reduce bias because different people can catch 'blind spots' that others might miss.

A. robotic
B. diverse
C. small
D. identical

Correct Answer: B. diverse

Explanation: Having people with different backgrounds, cultures, and viewpoints ensures that more potential issues are spotted before the AI is finished. Diversity helps uncover blind spots.

Q6 [Type: True/False | Difficulty: Easy | QID:57546]
True or False: AI is biased because it has feelings and intentionally wants to be unfair to certain groups.

A. True
B. False

Correct Answer: B. False

Explanation: AI has no feelings, opinions, or intentions. It is simply a pattern finder that mirrors the biases found in the data or instructions it is given. AI does not intentionally want to be unfair.

Q7 [Type: True/False | Difficulty: Easy-Medium | QID:57547]
True or False: AI bias is just a one-time 'glitch' that happens by accident and doesn't repeat.

A. False
B. True

Correct Answer: A. False

Explanation: Bias is defined as a systematic, repeating pattern where the AI is consistently wrong for a specific group of people over and over. It is not a random glitch.

Q8 [Type: True/False | Difficulty: Easy-Medium | QID:57548]
True or False: An AI algorithm can lead to unfair results even if the people who built it did not have any bad intentions.

A. True
B. False

Correct Answer: A. True

Explanation: Even with good intentions, designers can set goals (like maximizing watch time) that accidentally reward or amplify harmful or biased content. Bias can happen without bad intentions.

Q9 [Type: True/False | Difficulty: Easy | QID:57549]
True or False: Testing an AI system on many different groups of people is a way to detect bias before it causes real harm.

A. False
B. True

Correct Answer: B. True

Explanation: Running the AI on diverse populations allows auditors to compare results and see if the system works equally well for everyone. Testing is a key detection method.

Q10 [Type: True/False | Difficulty: Easy | QID:57551]
True or False: Once an AI model is launched and working, builders no longer need to check it for fairness.

A. True
B. False

Correct Answer: B. False

Explanation: Bias can appear at any stage, so it is critical to test and audit the outcomes both before and after a project is launched. Fairness is an ongoing responsibility.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:57552]
Which of these is a real-world consequence of AI bias in the healthcare industry?

A. Determining which hospital has the fastest Wi-Fi.
B. Changing the color of the doctor's scrubs.
C. Choosing the background music for the waiting room.
D. Deciding who gets medical treatment or an accurate diagnosis.

Correct Answer: D. Deciding who gets medical treatment or an accurate diagnosis.

Explanation: Bias in medical AI can lead to some groups receiving lower quality care or being overlooked for necessary treatments. This is a serious real-world consequence.

Q12 [Type: MCQ | Difficulty: Easy-Medium | QID:57553]
The YouTube algorithm sometimes shows angry or extreme videos because its main goal is to:

A. Delete all videos that contain arguments.
B. Keep people watching for as long as possible.
C. Make sure everyone gets equal time on screen.
D. Teach people how to be kinder to others.

Correct Answer: B. Keep people watching for as long as possible.

Explanation: The algorithm is optimized for watch time, and unfortunately, provocative or angry content often hooks viewers more effectively, leading to bias toward extreme videos.

Q13 [Type: MCQ | Difficulty: Medium | QID:57555]
What did the 'Gender Shades' study discover about major face-recognition AI tools?

A. They could only recognize people wearing sunglasses.
B. They worked perfectly for every person regardless of skin tone.
C. They were used to tag different types of shades of paint.
D. They failed much more often for dark-skinned women.

Correct Answer: D. They failed much more often for dark-skinned women.

Explanation: Researcher Joy Buolamwini found that error rates were as high as 34% for dark-skinned women compared to less than 1% for light-skinned men.

Q14 [Type: MCQ | Difficulty: Easy-Medium | QID:57556]
What is an 'AI Audit'?

A. A formal review to check if an AI system is biased or unfair.
B. A special type of battery used to power AI servers.
C. A way to make an AI run faster on old computers.
D. A process for charging users more money for AI services.

Correct Answer: A. A formal review to check if an AI system is biased or unfair.

Explanation: An audit involves testing the data, the algorithm, and the outcomes to ensure the system is working safely and fairly, much like a safety inspection for a building.

Q15 [Type: MCQ | Difficulty: Medium | QID:57557]
In the COMPAS case study, what was the AI tool trying to predict?

A. What the weather would be like outside the courthouse.
B. Which person would win a reality TV show.
C. The best route for a police officer to drive to work.
D. If someone might commit another crime in the future.

Correct Answer: D. If someone might commit another crime in the future.

Explanation: The COMPAS tool was used by courts to predict risk of reoffending, which influenced decisions about bail, parole, and jail time. It was found to be biased against certain groups.

Q16 [Type: MCQ | Difficulty: Easy-Medium | QID:57558]
Why is 'Human-in-the-Loop' an important strategy for reducing AI bias?

A. It ensures a person reviews AI decisions that affect people's lives.
B. It makes the AI code much shorter and easier to read.
C. It allows the AI to run without needing any electricity.
D. It turns the AI into a physical robot with human skin.

Correct Answer: A. It ensures a person reviews AI decisions that affect people's lives.

Explanation: Since AI does not understand context or fairness, having a human double check important outputs prevents the AI from making unfair automatic mistakes.

Q17 [Type: MCQ | Difficulty: Easy-Medium | QID:57559]
Which door does bias 'sneak through' when the people building the AI use their own unconscious assumptions to label data?

A. Data Bias
B. Human Bias
C. Hardware Bias
D. Algorithm Bias

Correct Answer: B. Human Bias

Explanation: When creators bring their own personal viewpoints or stereotypes into the build process, it is categorized as human-sourced bias. This can happen even without bad intentions.

Q18 [Type: MCQ | Difficulty: Easy-Medium | QID:57560]
A voice assistant that only understands American accents because it was mostly trained on audio from American adults is an example of:

A. Algorithm Bias
B. Human Bias
C. Language Bias
D. Data Bias

Correct Answer: D. Data Bias

Explanation: The system struggles with other accents simply because those voices were missing from its training library. This is a classic example of data bias due to lack of diversity.

Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:57561]
If an AI for grading school essays struggles with students who are non-native English speakers, who is being harmed by this bias?

A. The teachers who don't have to grade anymore.
B. The computer used to run the grading software.
C. Students who already have perfect grades.
D. Students learning English as a second language.

Correct Answer: D. Students learning English as a second language.

Explanation: If the AI only knows standard English patterns, it might unfairly give lower grades to students who write with different but valid structures, harming non-native speakers.

Q20 [Type: MCQ | Difficulty: Easy-Medium | QID:57562]
What is the most effective thing YOU can do if you notice an AI tool giving results that feel unfair?

A. Speak up and ask questions about why it's happening.
B. Try to fix the AI's raw code yourself at home.
C. Stop using all technology forever.
D. Ignore it because AI is always right.

Correct Answer: A. Speak up and ask questions about why it's happening.

Explanation: Questioning and sharing stories of AI failure helps draw attention to the problem so engineers can notice and fix it. AI literacy includes speaking up when something feels wrong.